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Fixed rate credit,
one swap away

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1 Abstract

Moonglade makes Morpho Midnight's fixed-rate loans tradable in a single swap. It is a bond desk on 1inch Aqua and SwapVM: lenders buy **mgNOTE**, an ERC-20 share of Midnight credit, with USDC and sell it back to exit; borrowers sell mgNOTE they do not yet hold, and the swap itself opens a fixed-rate loan on their own Midnight position.

A maker runs the desk as one Aqua strategy, priced by a custom SwapVM opcode that pairs the maker's own pricing curve with Midnight's tick grid. Capital never leaves the maker's wallet and inventory is minted at the moment of the trade, so the desk quotes both sides continuously and every trade settles as a real Midnight loan. Contracts are ownerless and immutable.

Three things are new: a SwapVM opcode that pairs any pricing contract with a lending protocol's tick grid; borrowing as a side effect of Aqua settlement, with no prefunded inventory and no borrower-side contract; and an ERC-20 that turns Midnight credit into composable collateral.

2 Motivation

Midnight's loans are fixed-rate and fixed-term, and illiquid until maturity. Its order book sets fixed rates but provides no liquidity: every trade needs a counterparty who has already posted an offer at that price. Offers do fill, but fills can be partial or delayed, and thin markets such as RWA collateral may have few resting offers at any moment.

- **Lenders** can enter or exit at the rate and size a resting offer provides; between offers, or in size, there is no continuous secondary market for their credit.
- **Borrowers** can read the book, but when nothing matches their size or rate they post an offer at their own rate and wait; what they cannot know is whether and when it fills.
- **Market makers** on Midnight are really lenders and borrowers running a carry trade: they earn only the spread between the offers they post, never a fee for providing liquidity, and they run bots to sign, refresh and cancel those offers.
- **Builders** cannot compose with Midnight credit. A position is a ledger entry, not an ERC-20, so it cannot be listed on a DEX, posted as collateral on Morpho Blue or Aave, wrapped into a vault, or laddered across maturities the way Pendle PTs or ERC-4626 shares can.

Midnight already has borrowers and distribution. What it lacks is the AMM half of its market.

3 Protocol overview

Every action is one MoongladeRouter.swap() against a desk on Aqua. The desk quotes a bid and an ask at every moment, so lending and borrowing rates are discovered continuously and the user never waits for a counterparty.

ACTION	SWAP	WHAT HAPPENS ON MIDNIGHT
Lend	USDC → mgNOTE	The desk's hook mints the mgNOTE it lacks from the maker's own position, just in time
Exit early	mgNOTE → USDC	The desk keeps the shares as inventory or burns them to cancel its own debt
Borrow	mgNOTE → USDC, holding none	The desk's hook mints the shortfall under a credit line the borrower granted, opening a fixed-rate loan on the borrower's collateral
Redeem	burn after maturity	Shares are burned for their part of the repaid loan

Three mechanisms make this work:

1. **mgNOTE** is a permissionless ERC-20 wrapper over one Midnight market's net credit. Mint by lending or borrowing, burn by repaying, redeem at maturity. One note per market, deployed deterministically by a factory.

2. MoongladeSwap is a new SwapVM opcode. It asks a maker-chosen curve for a quote, snaps the price to Midnight's tick grid with maker-favoured rounding, then validates the snapped amounts. The curve is a separate contract the maker ships; Moonglade is curve-agnostic, and the opcode's safety checks hold whatever the curve returns.
3. **The desk hook** keeps the maker's assets in their wallet and uses Aqua's preTransferOut and preTransferIn callbacks to mint or burn inventory inside settlement, so borrowing becomes a side effect of a swap.

Arbitrageurs connect the desk to Midnight's book through wrapFromOffer and unwrapToOffer, so the desk's rate cannot drift far from where loans actually clear.

4 Architecture

A swap enters the router, which prices it with the maker's curve and snaps it to Midnight's grid; Aqua settles it from the maker's wallet and calls the desk hook, which mints or burns mgNOTE against Midnight so inventory exists exactly when the swap needs it. Four contracts are Moonglade's; Aqua, SwapVM and Midnight are used as deployed.

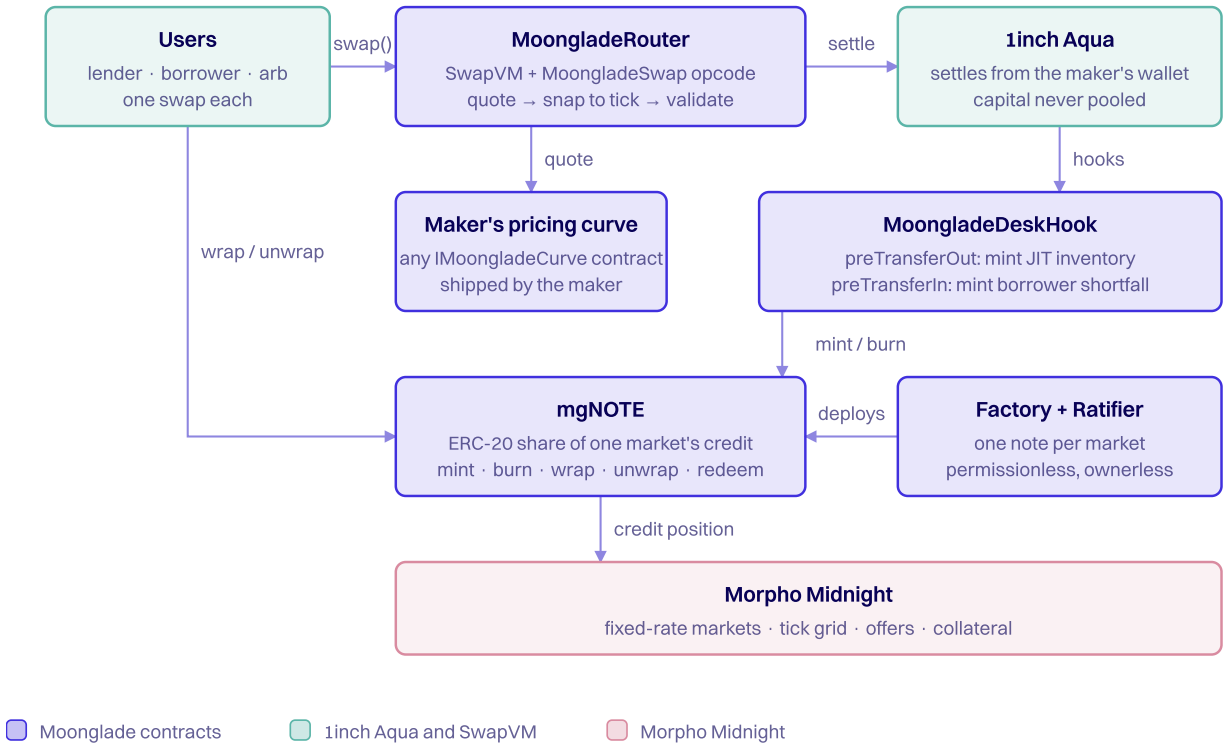


Figure 1. Moonglade architecture

5 Swap flows

Both directions are the same swap; the direction decides which hook fires and whose Midnight position the mint lands on.

5.1 Lend: USDC in, mgNOTE out

The maker never prefunds mgNOTE: the hook mints exactly the shortfall, against the maker's own credit or a new loan, inside the settlement that needs it.

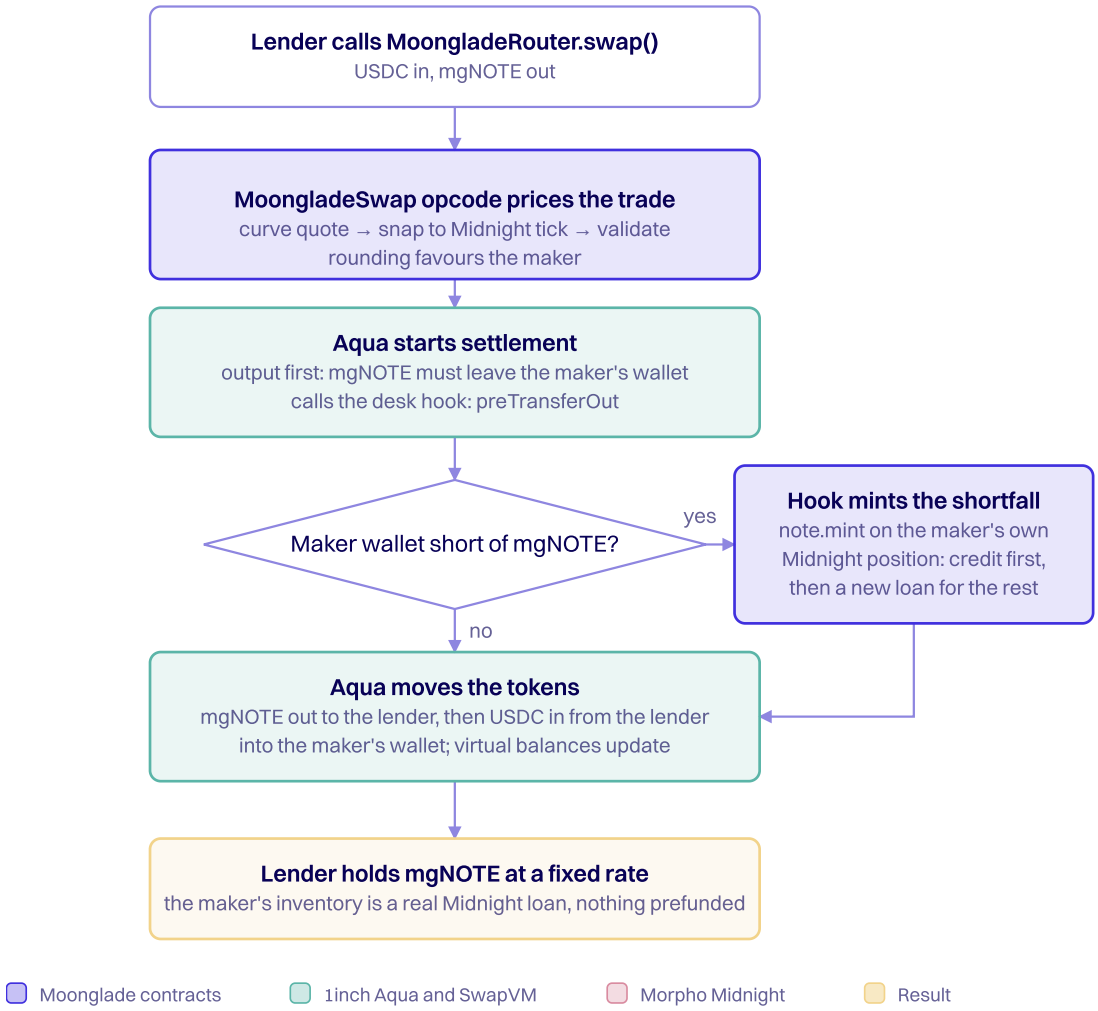


Figure 2. Lend flow

5.2 Borrow or exit: mgNOTE in, USDC out

An exiting lender and a borrower take the identical path; the only difference is that the borrower's shortfall is minted on the borrower's position, under a credit line it can revoke at any time.

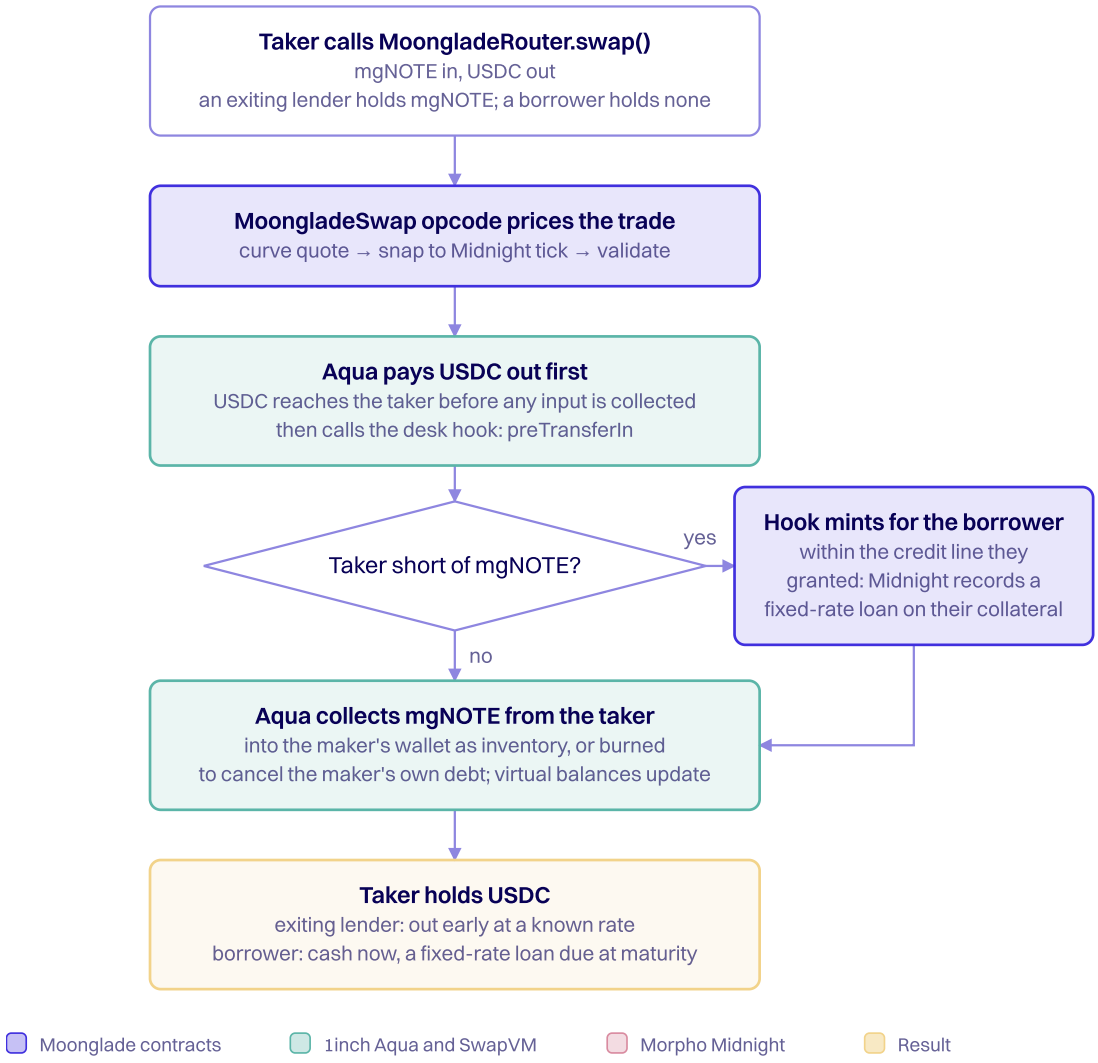


Figure 3. Borrow or exit flow

6 Economics and composability

The maker is a temporary counterparty on both sides of the market, not a lender of last resort. Its goal is never to run negative carry: it keeps exposure across lenders and borrowers near flat, takes positive carry when the spread offers it, and collects a fee on every swap. The flattening is done by arbitrage, not by the maker: when the desk's rate moves away from Midnight's book, arbitrageurs trade the gap through wrapFromOffer and unwrapToOffer and return the maker's inventory toward balance.

- **Profit = spread + fees.** Spread is the carry on inventory held between a lend and the borrow that offsets it; it can be positive or negative. Fees are charged on every swap and are the maker's steady income.
- **Net-zero by design.** Shares bought from a lender are handed to the next borrower and vice versa. When one side runs ahead, the curve moves the rate until the other side refills, so the maker's long or short exposure stays bounded between trades.
- **Fees above spot AMMs.** Users pay to lock in a rate and for instant liquidity on both sides of a credit market, which no order book offers; a fixed-rate swap can therefore carry a higher fee than a spot swap without losing flow.
- **Continuous discovery.** The rate is live at every block, so lenders and borrowers see the clearing rate before they commit rather than after an offer fills.

Beyond the desk, mgNOTE makes Midnight credit units composable and transferable across DeFi, as Pendle's PTs did for yield-bearing tokens: it can be listed, held, used as collateral in another lending market, or built into ladders, vaults and structured products, with every unit backed by a real Morpho loan.

7 Related work

Notional, Pendle and Timeswap each price a maturity-bearing token with a curve of their own, hold liquidity in a pool, and trade a token their own protocol issues. Moonglade keeps the curve and changes the venue: makers quote from their own wallets through Aqua, inventory is minted at the moment of the trade, and every swap settles as a real Morpho loan. Midnight's own order book remains the price reference; Moonglade is the market maker beside it. That is a new way of making markets on a lending protocol: the maker's inventory is debt it creates and cancels inside each swap, not tokens it holds.

8 Roadmap

Go-to-market is Midnight itself: every lender, borrower and maker already on it has one of the problems above today, so Moonglade needs no new market to bootstrap.

1. **v0, ETHGlobal Mumbai 2026.** One desk, one USDC market on a Base fork; lend, borrow, exit, arbitrage and redemption on-chain; SwapVM's invariant suite passing on the opcode.
2. **v1.** Swap fees, a collateral keeper for makers, and reference pricing curves with certified numerical bounds.
3. **v2.** mgNOTE as collateral on a Morpho Blue market (leveraged fixed yield), maturity ladders and fixed-yield vaults built on the token, and aggregator routing across the desk and Midnight's book.

9 Author and licence

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